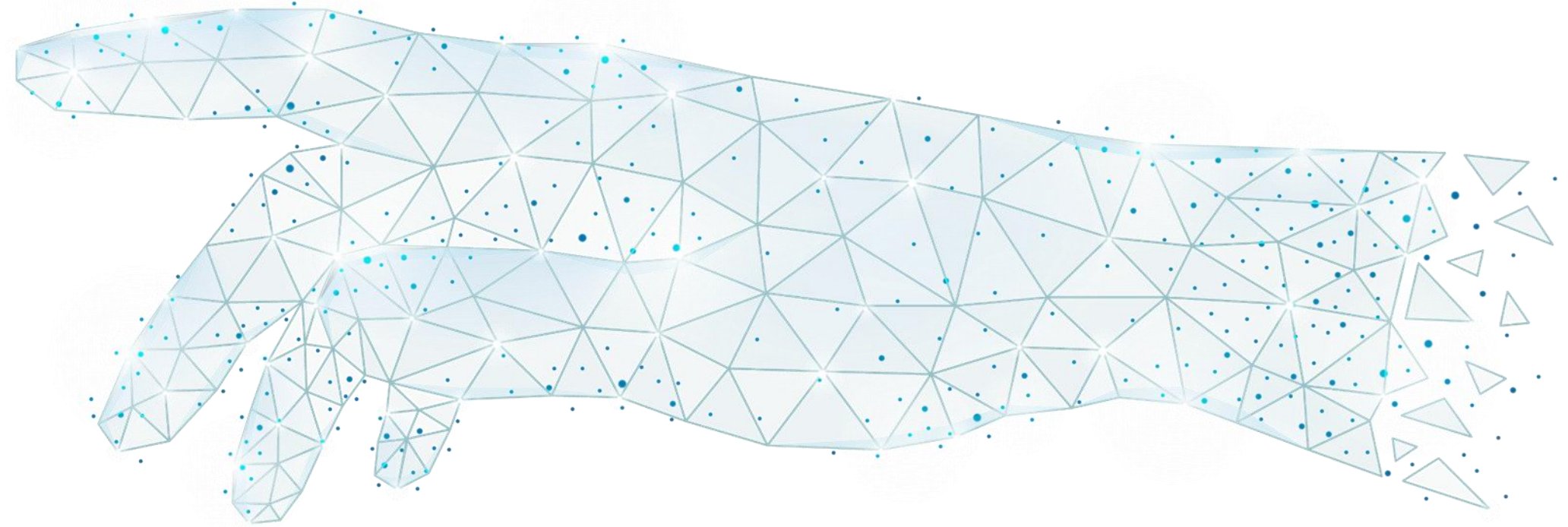


BUSINESS PROPOSAL FOR CAAM



01

Executive Summary



CURRENT CHALLENGE



ILLEGAL STRUCTURE/NO RECORD

1. NO AOL INSTALL TO STRUCTURE

NEW APPLICATION

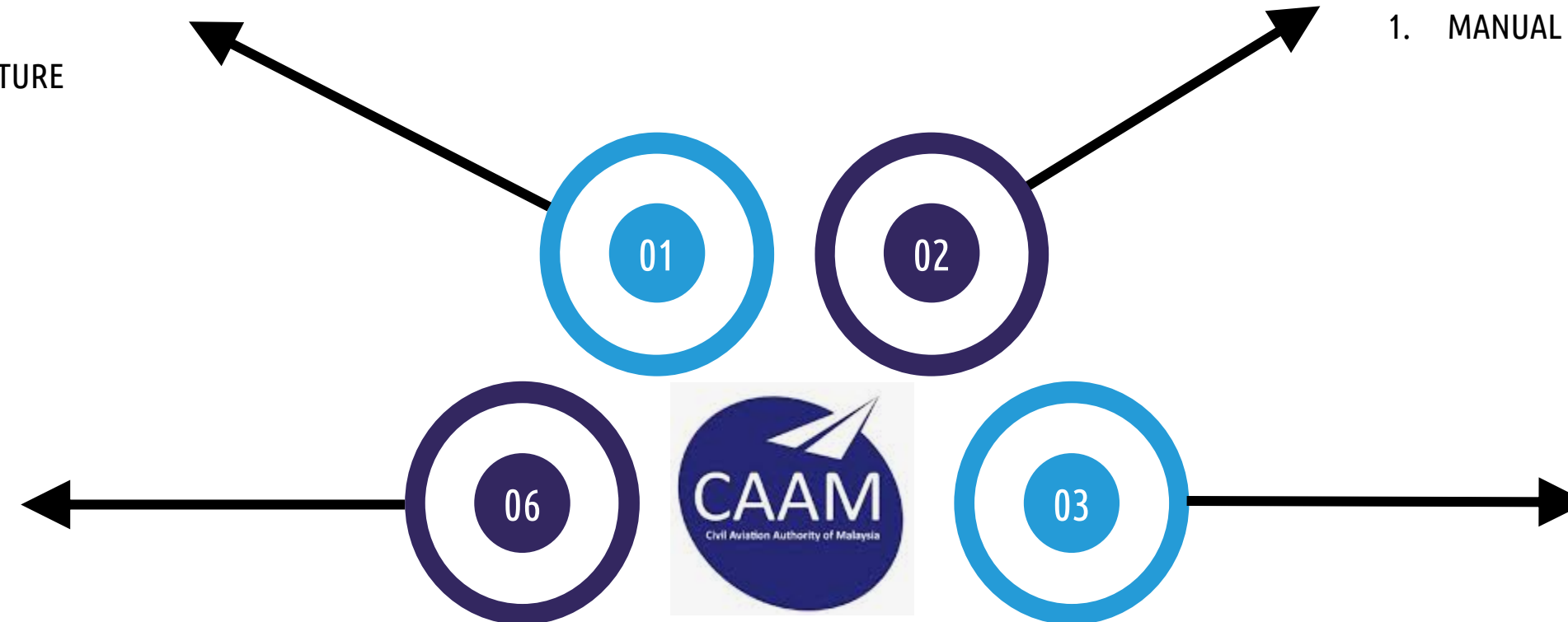
1. MANUAL METHOD (HARDCOPY)

AOL MALFUNCTION

1. AOL NOT PROPERLY FUNCTION
2. METHOD OF AUDIT VERY CONVENTIONAL (MANUAL)

LACK OF COMPLIANCES

1. INDUSTRY TAKE FOR GRANTED ON CAAM REQUIREMENT AND ACT



EXECUTIVE SUMMARY



E-Application portal

- The E-Applications System is online platform that allows users to submit application digitally
- It replaces paper forms with a secure, easy-to-use web interfaces

Key feature

- Online form submission
- Document upload
- Application status tracking
- Automated email or SMS notifications
- Admin panel for reviewing and managing application

Benefits

- Saves time and reduces manual work
- Fewer errors and more accurate data
- Accessible from anywhere, anytime
- Supports eco-friendly, paperless process
- Offers data insight through reporting tools

Regulation & ACT



Civil Aviation Act 1969 (Act 3)

Civil Aviation Regulations 2016

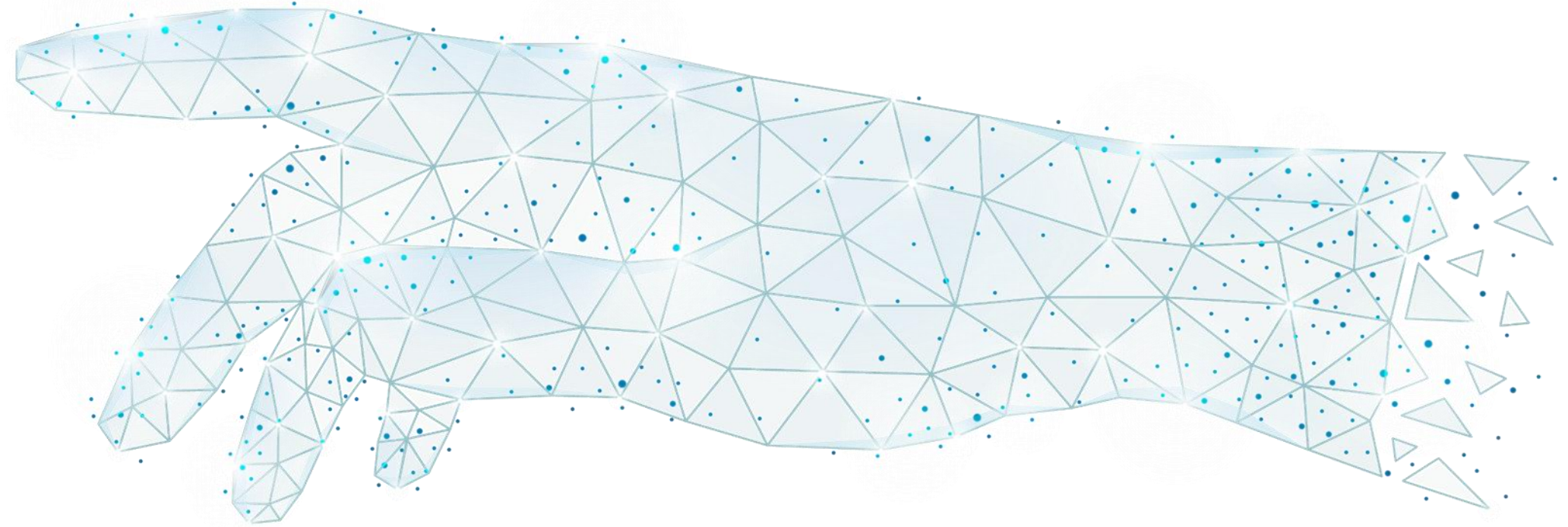
CAAM CAD Volume 14

Airport Standards Directive 404(ASD 404):Lighting of Obstacle

Section 240 of the Civil Aviation Act 1969 (Act 3)

02

Problem Statement

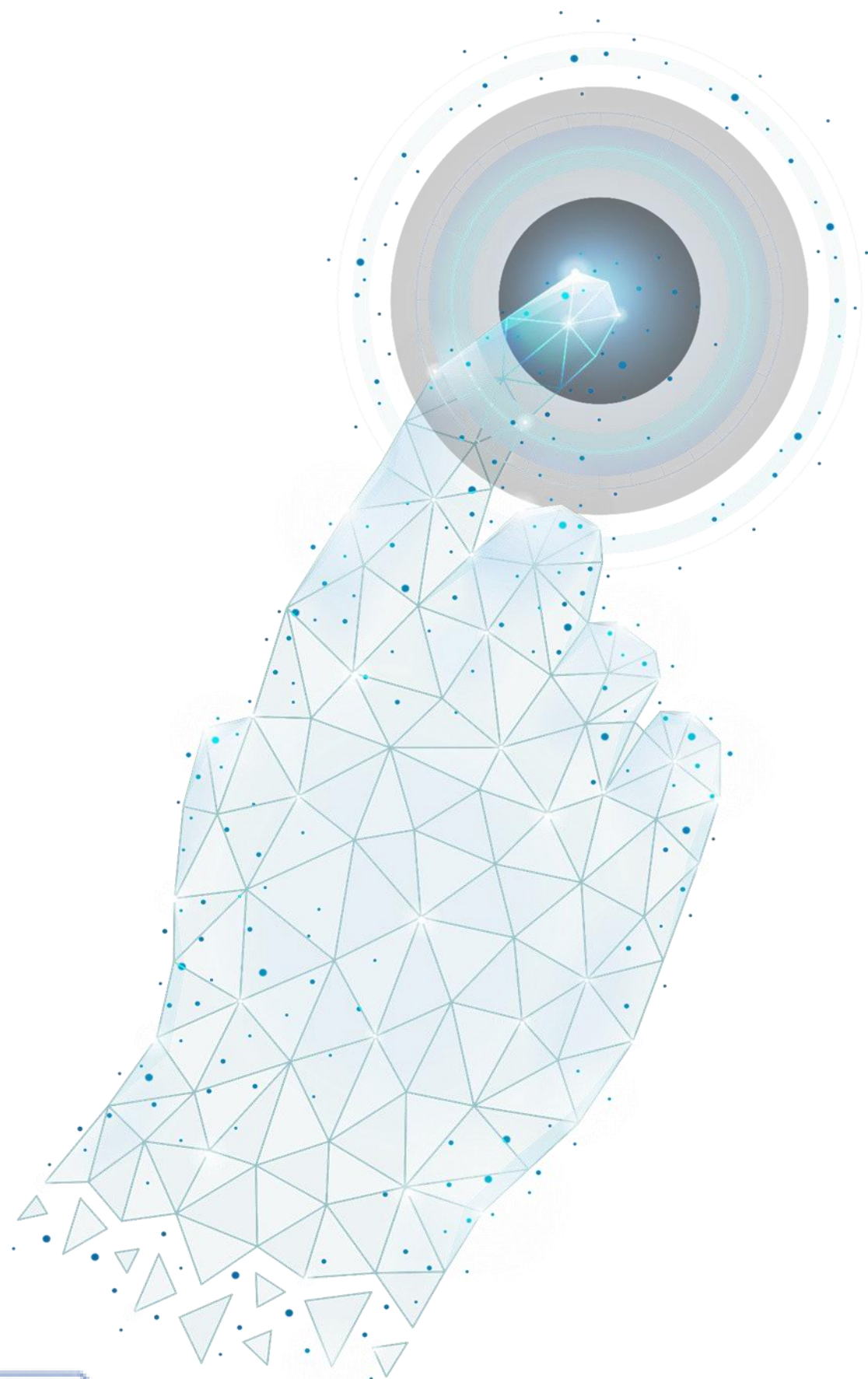


Problem

1. **Traditional paper-based systems** leading to delays, errors, and difficulty in tracking.
2. **No centralized digital platform** to handle new applications, license renewals, or compliance monitoring.
3. **Unregistered and unlicensed structures** continue to operate without Aviation Obstruction Lights (AOL) posing a safety risk.

Solution

1. Introducing E-application system for CAAM for all new applications and renewals.
2. User-friendly, secure, and scalable web-based interface, allowing applicants to submit and manage their applications online efficiently.
3. Digitalize form submission, real-time application tracking, automated notifications, secure document uploads, and backend administrative tools for reviewing and processing applications.



03

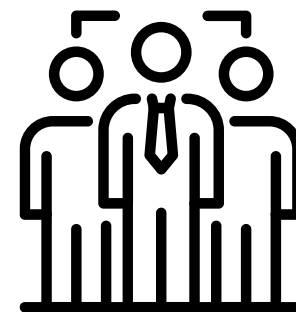
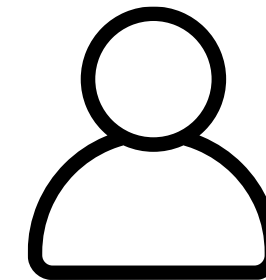
Objective

OBJECTIVES



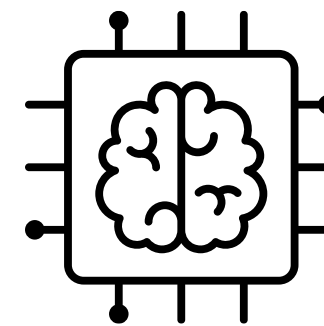
Digitalize all application for approval of height limitation through a centralized online platform.

Enable **new applications & renewal** with compulsory AOL requirements based on structure.



Provide a real-time system for AOL tracking and monitoring.

Compliance to Act - Support the **legalization** of unregistered structures and AOL installations



TYPES OF STRUCTURE REQUIRE AOL



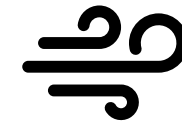
01

MOBILE
NETWORK
OPERATOR
TOWER



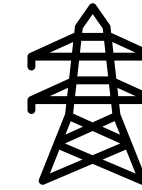
02

HIGH RISE
BUILDINGS



03

WIND MAST
MEASUREMENT



04

TRANSMISSION
LINE



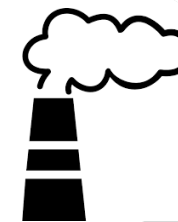
06

CRANE



07

RADAR



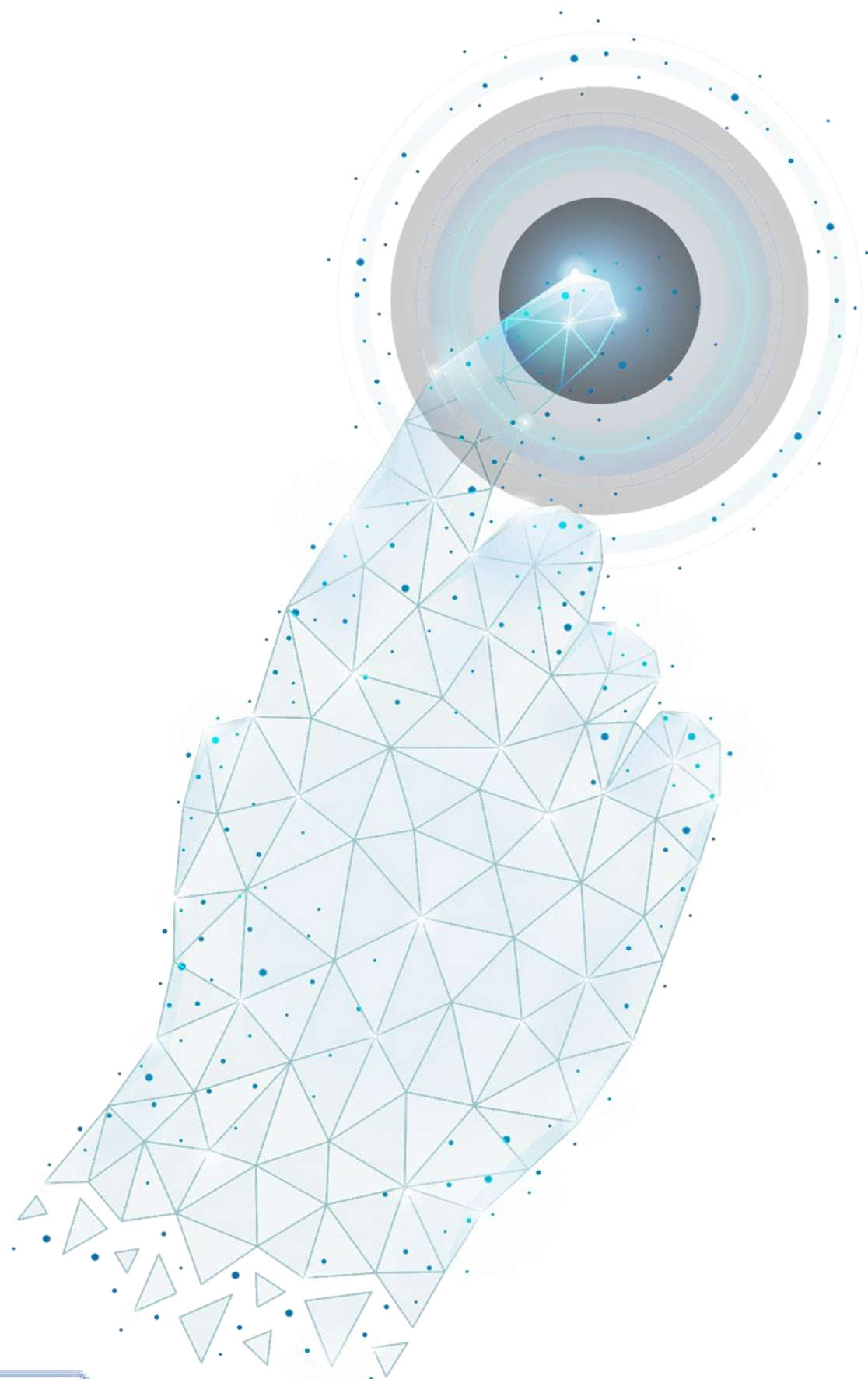
08

CHIMNEY



09

HELIPORT/
HELIDECK



04

Benefits



BENEFITS TO CAAM



INCREASE REVENUE

1. REGISTRATION (GREENFIELD)
2. YEARLY RENEWAL (BROWNFIELD)

RISK MITIGATION

1. FREQUENTLY UPDATE ON THE AOL CONDITIONS AND FUNCTIONALITY

DATA UPDATE

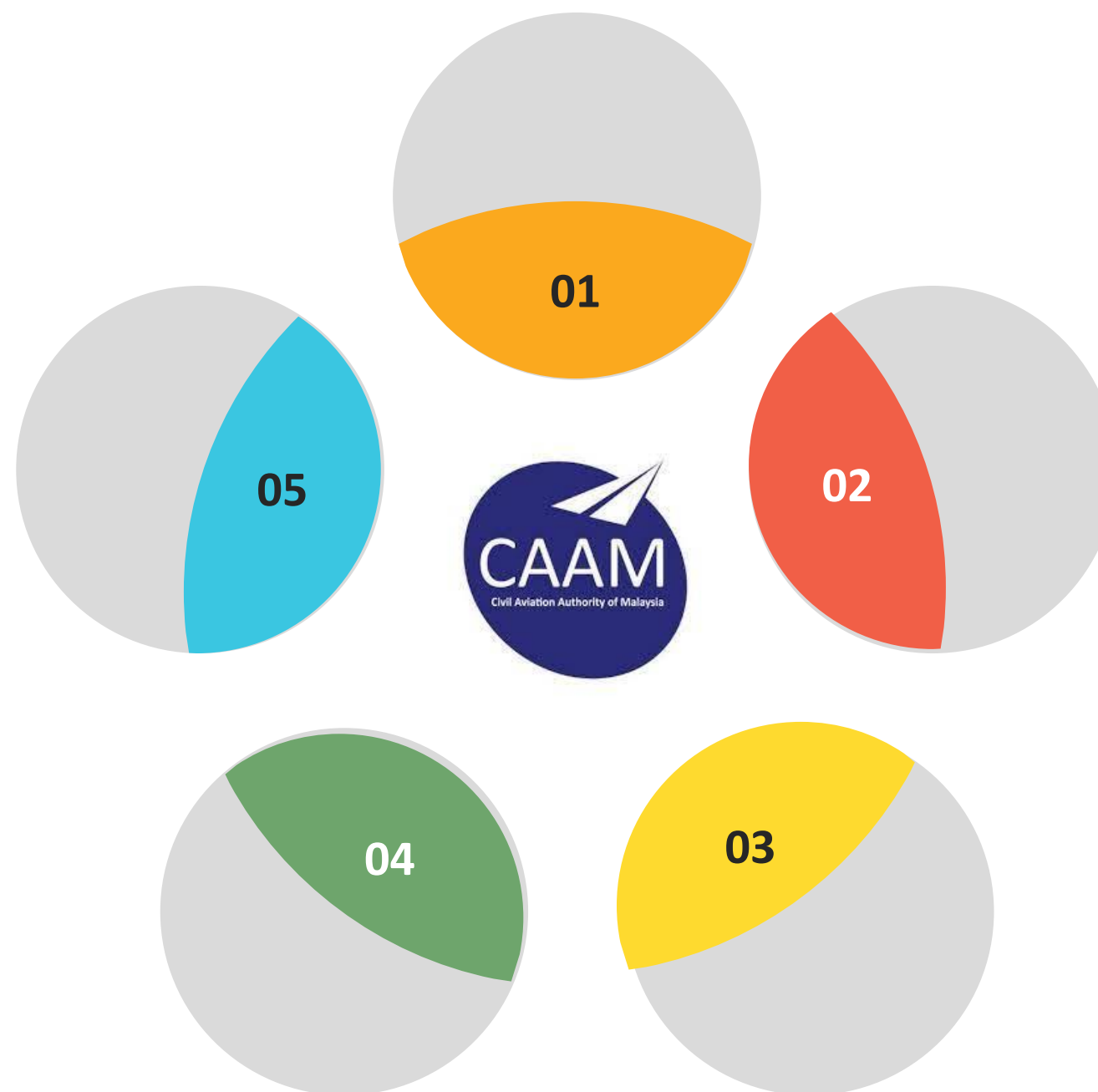
1. DATA UPDATE FOR FUTURE DEVELOPMENT

DIGITALIZATION

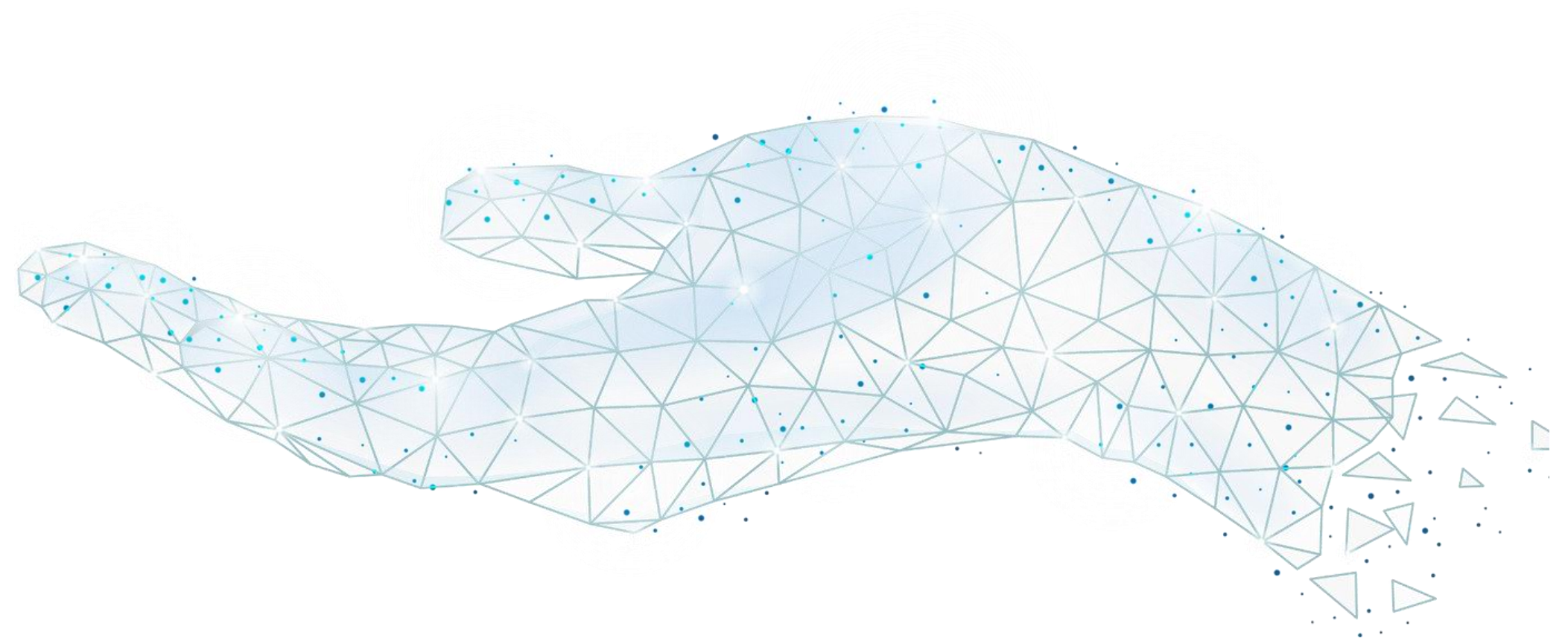
1. ALIGN WITH THE LATEST TECHNOLOGY

LEGAL COMPLIANCES

1. STRUCTURE LEGALIZATION (PEMUTIHAN) PROGRAM



05



Proposed Solutions: E-application System

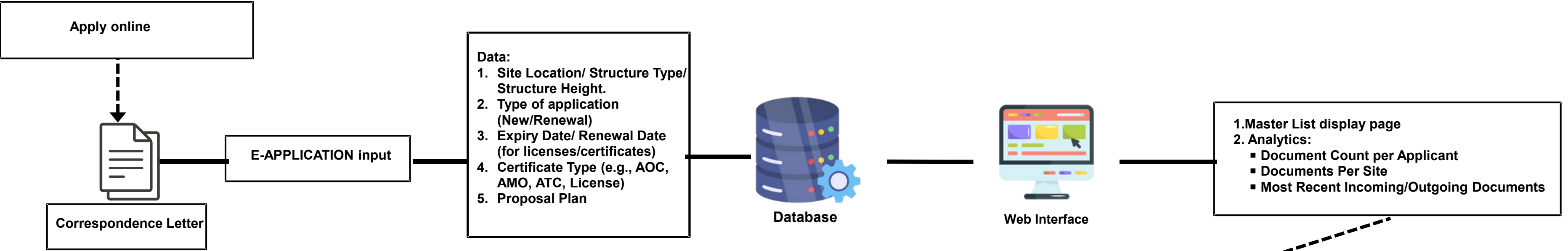
COMPONENTS



Task Category	Task Name
Project Management	• Requirement Gathering • Finalizing Scope of Work (SOW)
Art & UI/UX	• Concept Mock-ups • UI/UX and Art Assets Creation
Frontend Development	• Web-based dashboard (Based on mock-ups. • App publishing/Online System

Task Category	Task Name
Backend Development	• Core Engine (Nextjs) • Features Development • Correspondence Letter, Data Site • Multiple User Account • Login System
Security & Database	• Security Features • Reporting System • Server Hosting & Configuration

ARCHITECTURE PLAN

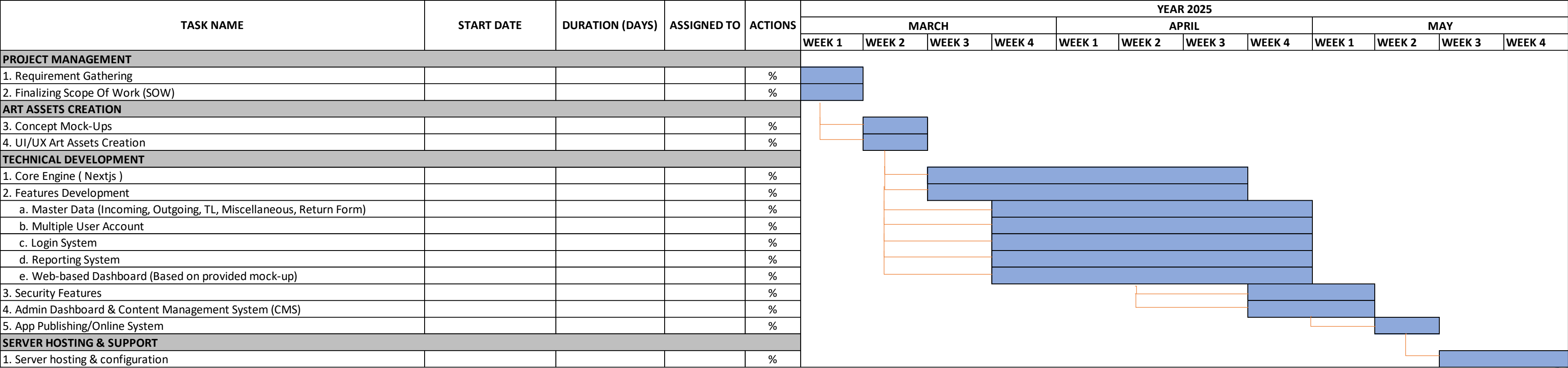


Navigation: All regions, All types, All client, Date Issue, Date Received

Reference Number	Client	Date Issue	Date Receive	Region	Type	
test doc 5	E EDOTCO	March 20th, 2025	March 29th, 2025	Southern	Outgoing	⋮
test doc 4	A ALPHA ORANGE	March 20th, 2025	February 22nd, 2025	Eastern	Outgoing	⋮
test doc 3	A ALPHA ORANGE	March 4th, 2025	March 18th, 2025	Southern	Incoming	⋮
test doc 2	A ALPHA ORANGE	February 5th, 2025	March 12th, 2025	Central	Incoming	⋮
test doc	E EDOTCO	March 1st, 2025	February 13th, 2025	Central	Incoming	⋮

Previous Next

DEVELOPMENT TIMELINE



DEVELOPMENT TOOLS



Frontend Development

- Next.js
- React
- TailwindCSS
- ShadCN UI

Backend & Database

- NextJs
- MySQL

Code Editor

- Cursor
- Visual Studio Code

Version Control

- Git

UI/UX ART ASSET CREATION



LOGIN PAGE

Create Account

Sign up

Create Account

Already have account ? [Sign in](#)

Welcome Back

Sign in

Login

Don't have account ? [Sign up](#)

UI/UX ART ASSET CREATION



HOME PAGE

[Home](#)

Master List

Clients

Sites

[Home](#)

Total Incoming Letter

—

Total Outgoing Letter

—

Total Client

—

Total Site

—

Recent Activity

[illegible]

CAAM
Civil Aviation Authority of Malaysia

Sites

4 New

[illegible]

CREATE FORM PAGE



Create Site

Type of Document ▼

Client

 Date Issued

 Date Received

Reference Number

Title

Local Council

Project ▼

Type ▼

Site ID

Site Name

Region

Save Changes

EDIT FORM PAGE



Edit Site

Type of Document ▼

Client



Date Issued



Date Received

Reference Number

Title

Local Council

Project ▼

Type ▼

Site ID

Site Name

Region

Save Changes

VIEW DETAIL PAGE



[Home](#) [Master List](#) [Correspondence](#)

Correspondence >

Incoming Letter

Site ID: XXX

Site Name: ABC SDN BHD

Local Council: XXX

Reference Number: XXX

Date Issued: 11/9/2024

Date Received: --

Title:

Upload

Site ID: XXX

Site Name: ABC SDN BHD

Local Council: XXX

Reference Number: XXX

Date Issued: 12/9/2024

Date Received: --

Title:

Upload

Outgoing Letter

Site ID: XXX

Site Name: ABC SDN BHD

Local Council: XXX

Reference Number: XXX

Date Issued: 2/8/2024

Date Received: 6/9/2024

Title:

Upload

Site ID: XXX

Site Name: ABC SDN BHD

Local Council: XXX

Reference Number: XXX

Date Issued: 28/10/2024

Date Received: 28/10/2024

Title:

Upload

Site ID: XXX

Site Name: ABC SDN BHD

Local Council: XXX

Reference Number: XXX

Date Issued: 2/8/2024

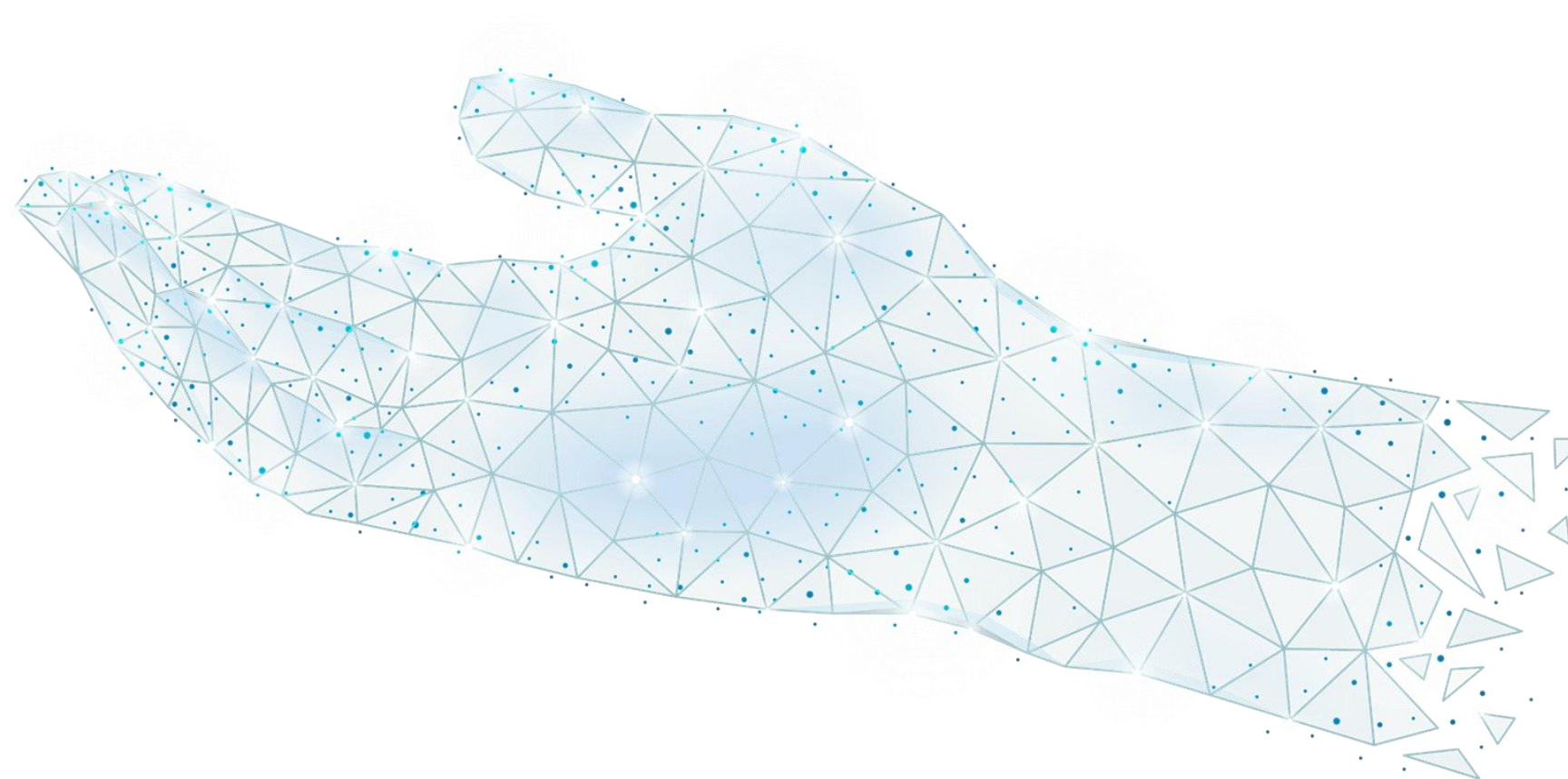
Date Received: 20/8/2024

Title:

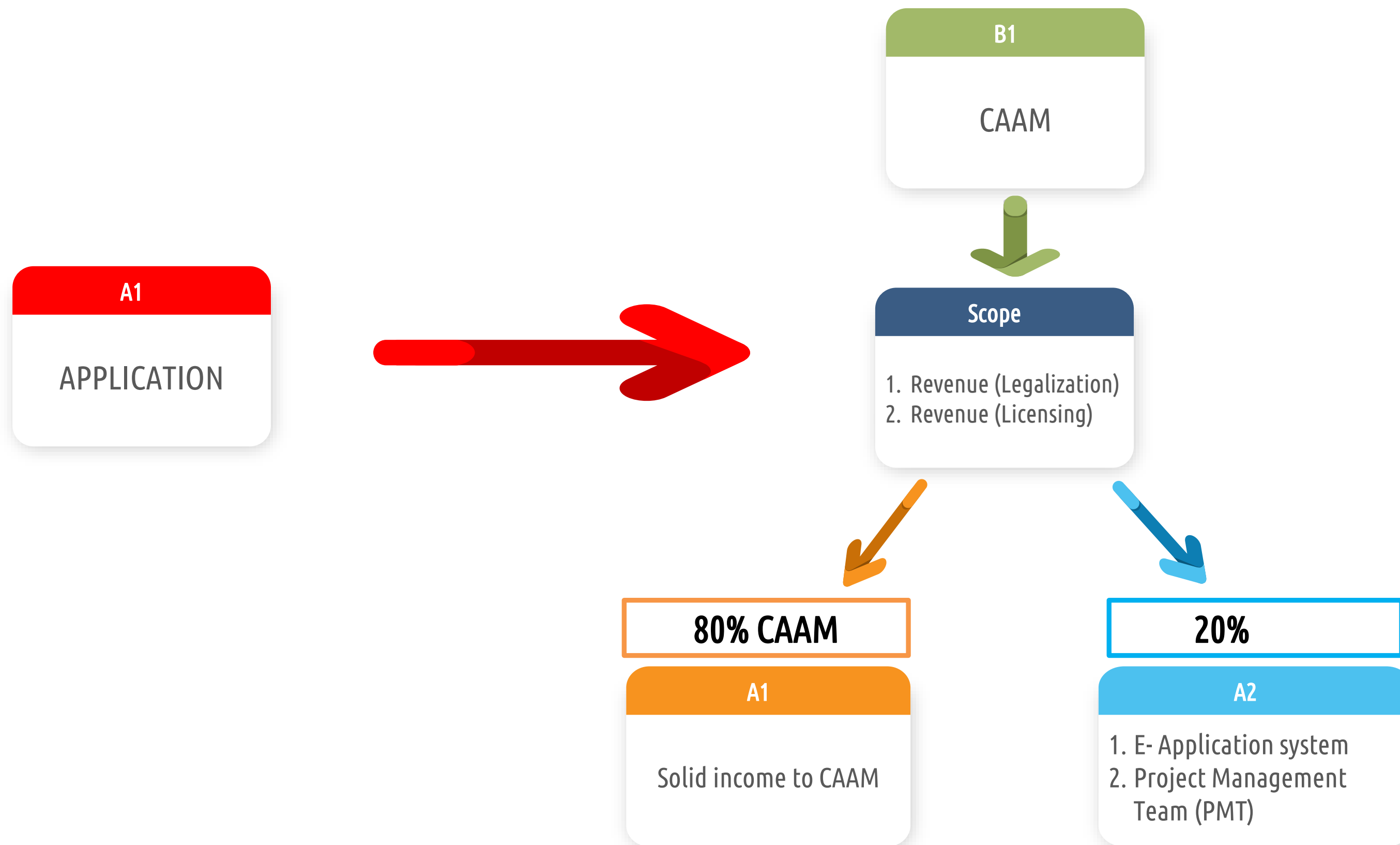
Upload

08

Commercial



PROFIT SHARING MODEL



REVENUE & PROFIT



Component	No of sites	Estimated (RM)
Revenue		
Legalization	2000	400,000
Licensing	5000	1,500,000
Total Revenue (per month)		1,900,000
Total Revenue (12 Months)		22,800,000
Cost (CAPEX & OPEX)		
E-application system	LS	200,000
PMT	LS	250,000
Miscellaneous	LS	81,000
Total Cost (per month)		531,000
Total Cost (12 Months)		6,372,000
Profit (per month)		1,369,000
Profit (12 Months)		16,428,000

Yearly Forecast Cashflow



Details Projection Income	Total Sites	Fee	Year-6	Year-7	Year-8	Year-9	Year-10
Telco Tower	60,000	RM2,000.00					
Structure	12,000	RM2,000.00					
High rise building	21,000	RM2,000.00					
Transmission Tower	210,000	RM500.00					
Cranes	12,000	RM2,000.00					
Aviation Warning Light	300,000	RM500.00					
Yearly income							
Telco Tower	60,000	RM500.00	30,000,000.00	30,000,000.00	30,000,000.00	30,000,000.00	30,000,000.00
Structure	12,000	RM500.00	6,000,000.00	6,000,000.00	6,000,000.00	6,000,000.00	6,000,000.00
High rise building	21,000	RM500.00	10,500,000.00	10,500,000.00	10,500,000.00	10,500,000.00	10,500,000.00
Transmission Tower	210,000	RM100.00	21,000,000.00	21,000,000.00	21,000,000.00	21,000,000.00	21,000,000.00
Cranes	12,000	RM500.00	6,000,000.00	6,000,000.00	6,000,000.00	6,000,000.00	6,000,000.00
Aviation Warning Light	300,000	RM300.00	90,000,000.00	90,000,000.00	90,000,000.00	90,000,000.00	90,000,000.00
Total payment received			RM163,500,000.00	RM163,500,000.00	RM163,500,000.00	RM163,500,000.00	RM163,500,000.00

Yearly Forecast Cashflow



Expenses							
Registration System (E-application)	Pay as services	RM200,000.00	RM2,400,000.00	RM2,400,000.00	RM2,400,000.00	RM2,400,000.00	RM2,400,000.00
PMT	Pay as services	RM250,000.00	RM3,000,000.00	RM3,000,000.00	RM3,000,000.00	RM3,000,000.00	RM3,000,000.00
Miscellaneous		RM1,141,500.00	RM1,155,000.00	RM1,155,000.00	RM1,155,000.00	RM1,155,000.00	RM1,155,000.00
Total expenses			RM6,555,000.00	RM6,555,000.00	RM6,555,000.00	RM6,555,000.00	RM6,555,000.00
Total ending bank balance			RM156,945,000.00	RM156,945,000.00	RM156,945,000.00	RM156,945,000.00	RM156,945,000.00



THANK YOU